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Aim Use sequence data types and their associated operations in Python a) Strings b) Lists c) Arrays d) Tuples e) Dictionary f) Sets g) Range

Theory 1 String a string is a sequence of characters enclosed in single (' ') or double (" ") quotes. immutable (cannot be changed after creation). supports indexing, slicing, and built-in method. example operator: len (), slicing, .lower (), .upper (). 1. lists a list is an ordered, mutable collection of elements. element can be different data type. more memory efficient than arrays for numeric data. ex operation: append (). remove (), insert (), max (). 3 Arrays Arrays store homogeneous (same type) elements. More memory efficient than list for numeric data ex operations: append (), pop () 4. tuple a tuple is an ordered, immutable collection of elements. define using parentheses (). faster than lists for fixed data. 5. Dictionary a dictionary stores data in key-value pairs. keys must be unique. mutable data structure ex operation: keys (), values (), items () 6. Sets A set is an unordered collection of unique element. automatically removes duplicates support mathematical operations like union, intersection. 7. Range range () generates a sequence of no. used in loops. Syntax: range (start, stop, step) Q1. What are sequence data types in python? Ans. Sequence data type store ordered collection of elements. types: string, list, Tuple, Range, Array Indexing: positive index > starts from 0 negative index > start from -1 example s = "python" print(s[0]) print(s[-1]) Slicing: Extract part of sequence. print(s[0:3]) Q2. Difference between List and Tuple Ans. list > mutable (can change), uses [] Tuple > immutable (cannot change), uses () why prefer tuple? faster more memory efficient protects data from modification can be used as dictionary keys Q3. Difference between List and Array Ans. List > Can store different data types. Array > Store same data type only. Use array when: working with large numeric data. Memory efficiency is required. Q4. How dictionaries work internally? Ans. Dictionaries use a hash table. • each key is converted into a hash value. -hash determine storage location. • lookup is fast (O(1) average time). Dictionary lookups are faster than lists because lists require linear search (O(n)). Q5. difference between set and List. Ans. List > ordered allows duplicates. Set > Unordered, no duplicates. Sets automatically remove duplicates: set ([1,2,2,3]) Practical Use: -Removing duplicates Membership testing - Mathematical operation (union, intersection)

```
string1 = " Artificial Intelligence"
print("String :", string1)
print("Length:", len(string1))
print("First Three characters:", string1[:3])
print("\n-----")
list1 = [10, 25, 5, 40, 15]
print("List:", list1)
print("Maximum:", max(list1))
print("Minimum:", min(list1))
print("\n-----")
student = ("shri", "ram", "sham", "rajshri")
print("Student Names:")
for name in student:
    print(name)
print("\n-----")
student_dict = {101:"shri", 102:"ram", 103:"rajshri"}
print("Dictionary:", student_dict)
print("Keys (Roll Number):")
for key in student_dict.keys():
    print(key)
print("\n-----")
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print("\n")
list_with_duplicate = [1,2,3,2,4,1,5]
unique_set = set(list_with_duplicate)
print("original List:", list_with_duplicate)
print("List after removing duplicate :", list(unique_set))

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String : Artificial Intelligence
Length: 24
First Three charcters:  Ar

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List: [10, 25, 5, 40, 15]
Maximum: 40
Minimum: 5

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```

-----
Student Names:
shri
ram
sham
rajshri

```

```

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Dictionary: {101: 'shri', 102: 'ram', 103: 'rajshri'}
Keys (Roll Number):
101
102
103

```

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original List: [1, 2, 3, 2, 4, 1, 5]
List after removing duplicate : [1, 2, 3, 4, 5]

```

conclusion :

In this practical , various sequence data types in Python were implemented successfully. We performed : - string operations(length, slicing) -List operation (max, min,insertion,deletion) Tuple traversal -Dictionary key access -Set for duplicate removal -Vowel counting logic -Type conversion between list and tuple -Range for iteration The program executed without errors and demonstrated proper understanding of sequence data structure and their associated operation in python